I have many engagement scores based on different features: RMS+Audio energy, spectral features, emotion interpretation (audio only), pitch + formants, transcript interpretation, and video salience; please think deeply on how I should weigh these features to compile into an overall engagement score and then please give me a table of these weights and alternative weighing based on interpretation

Based on extensive research into multimodal engagement detection and feature weighting methodologies, I'll provide you with a comprehensive weighting scheme for your engagement scoring system. The weights should reflect both the **reliability** and **discriminative power** of each feature type for engagement detection.

Recommended Weighting Scheme

Feature Category	Primary Weight	Alternative Weight	Rationale
Emotion Interpretation (Audio)	25%	30%	High correlation with engagement states; validated in literature [1] [2] [3]
Video Saliency	22%	18%	Strong visual attention indicator; proven engagement predictor [4] [5] [6]
Transcript Interpretation	20%	25%	Content understanding crucial for engagement; semantic analysis reliability [7] [8] [9]
RMS + Audio Energy	15%	12%	Reliable baseline measure; indicates vocal intensity and participation [10] [11] [12]
Spectral Features	12%	10%	Complementary to RMS; captures vocal quality variations [13] [14] [15]
Pitch + Formants	6%	5%	Useful but highly variable; cultural and individual differences [16] [17] [18]

Deep Analysis of Feature Importance

High-Weight Features (20%+)

Emotion Interpretation (Audio Only) receives the highest weight because emotional state is the most direct indicator of engagement. Research shows that emotion recognition from audio achieves 50-60% accuracy, which is comparable to human performance, making it highly reliable for engagement assessment.[1] [2]

Video Saliency captures visual attention patterns that strongly correlate with engagement. Modern saliency detection algorithms achieve high performance by integrating spatiotemporal cues and uncertainty weighting, making this a robust engagement indicator.[4] [6]

Transcript Interpretation provides semantic understanding of content engagement. Text analysis can reveal engagement through sentiment, topic relevance, and linguistic markers, with automated scoring showing strong correlation with manual evaluation.[7] [8] [9]

Medium-Weight Features (10-20%)

RMS + Audio Energy serves as a reliable baseline measure. While not as nuanced as emotion detection, energy levels consistently indicate participation and vocal engagement across different contexts.[10] [11]

Spectral Features complement energy-based measures by capturing vocal quality variations. These features are particularly useful for detecting subtle engagement changes that energy alone might miss.[13] [14]

Lower-Weight Features (<10%)

Pitch + Formants receive the lowest weight due to high individual variability and cultural differences. While formants can indicate emotional states, they require careful normalization and are less reliable across diverse populations.[16] [17] [18]

Alternative Weighting Interpretations

Content-Heavy Scenarios (Alternative Weights)

- **Transcript Interpretation: 25%** - When content comprehension is primary
- **Emotion Interpretation: 30%** - When emotional engagement is critical
- **Video Saliency: 18%** - Reduced when content is more important than visual attention

Presentation/Lecture Contexts

- Increase **Video Saliency** to 25% for visual attention tracking
- Reduce **Transcript** to 15% if content is predetermined
- Maintain high **Emotion** weighting for audience response

Interactive/Discussion Settings

- Increase **RMS + Energy** to 20% for participation measurement
- Increase **Pitch + Formants** to 10% for conversational dynamics
- Reduce **Video Saliency** to 18% as attention may be distributed

Implementation Recommendations

Temporal Weighting: Consider applying different weights across time windows - higher weights for **Emotion** and **Energy** during active discussion periods, higher **Saliency** weights during presentation phases. [19] [20]

Adaptive Weighting: Implement confidence-based weighting where features with higher certainty receive proportionally more influence. [21] [22]

Cultural Calibration: Adjust **Pitch + Formants** weights based on demographic data, as these features show significant cultural variation. [18]

This weighting scheme is grounded in research showing that multimodal fusion with appropriate feature weighting can achieve 8-9% performance improvements over single-modality systems, with optimal results coming from late fusion approaches that respect the unique characteristics of each feature type. [23] [22] [24] [19]

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